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Prereq: Lab 04: Deployment Rollback & Recovery

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CI/CD ENGINEERING · KNOWLEDGE BASE

Lab 05: Rollback

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Objective

Break production deliberately, then recover by redeploying the previous known-good digest — and write the postmortem.

Prerequisites

- Lab 04 green with two successful production deploys (you need a previous good digest); concepts: Rollback & Recovery.

Architecture

Bad merge → staging green-but-wrong (tests miss it) → approved to prod → incident → redeploy previous digest → postmortem with a pipeline action item.

Setup

Note the current production digest; keep its value handy.

Step 1 — Cause the incident

Execute

Merge a change that passes tests but breaks the app's visible output (e.g. wrong port, broken route). Promote to production.

Expected Result

Production is broken despite a green pipeline.

Why

Green ≠ correct: tests cover what you thought of. This is why recovery must be engineered, not improvised.

Step 2 — Roll back, don't rebuild

Execute

Redeploy the previous digest to production (re-run the old `deploy-prod` or `dispatch` with the saved digest). Time yourself.

Expected Result

Production healthy in minutes; no fresh build involved.

Why

Old digests are proven; fresh builds add untested variables to a fire (see rollback rule).

Step 3 — Postmortem with teeth

Execute

Write: timeline, root cause, and one action item that becomes a pipeline change (a new test, check, or alert) — then implement it.

Expected Result

A merged pipeline change that would have caught this incident.

Why

Postmortems without merged changes are diary entries.

Verification

- [] Recovery time measured (aim: minutes).
- [] Previous digest redeployed, not a fresh build.
- [] One preventive pipeline change merged.

Failure Scenarios

- Previous digest unknown: promotion didn't record digests — log every deployed digest from now on.
- Rollback also broken: data migrated forward — forward-fix with expand-contract instead.

Cleanup

Keep the postmortem in the repo (docs/postmortems/).

What This Proved

Recovery as a practiced path. General labs complete — Jenkins track starts at Lab 06.

Interview Notes

- Rollback vs forward-fix: which failure forces expand-contract instead of reverting.
- Why the deployed digest must be recorded — what "previous digest unknown" costs you mid-incident.

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